ValPay Portfolio 2: Query Optimisation

MIT 8103 Advanced Database Systems · Evidence

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Compiled from `portfolio-2-query-optimisation/evidence`: 9 screenshots in `screenshots/` and script output in `script-output/` (`explain_plans.txt`, `query_results.txt`). Text output is reproduced verbatim.

1. QUERY RESULTS

Q1. Wallet statement for a customer

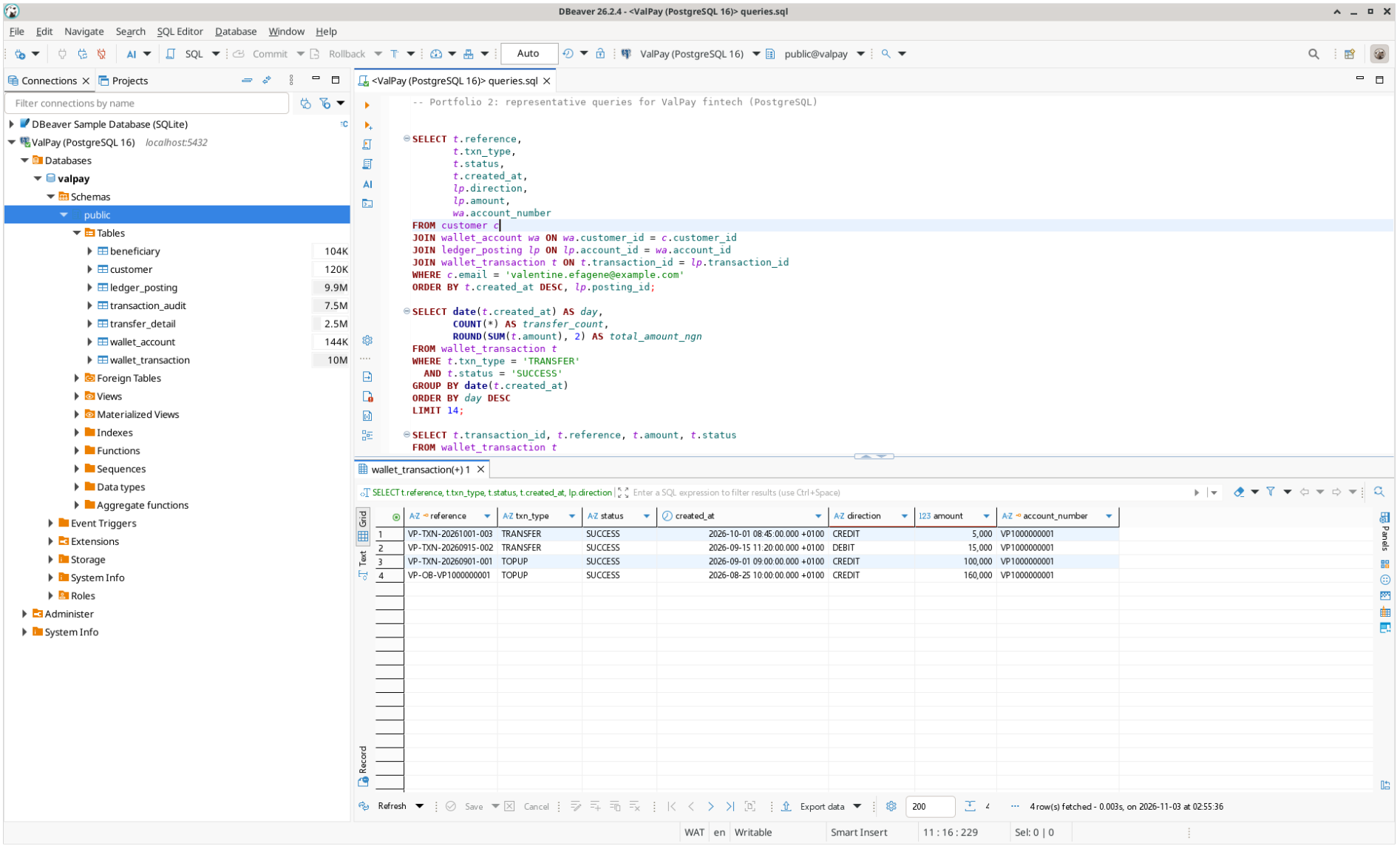


Figure 1. *Q1 in DBeaver: wallet statement for valentine.efagene@example.com, newest first.*

Source: evidence/script-output/query_results.txt

```
Border style is 2.
Output format is aligned.
=== Q1: Wallet statement for a customer (by email) ===
+---------------------+----------+---------+------------------------+-----------+-----------+----------------+
|      reference      | txn_type | status  |       created_at       | direction |  amount   | account_number |
+---------------------+----------+---------+------------------------+-----------+-----------+----------------+
| VP-TXN-20261001-003 | TRANSFER | SUCCESS | 2026-10-01 08:45:00+00 | CREDIT    |   5000.00 | VP1000000001   |
| VP-TXN-20260915-002 | TRANSFER | SUCCESS | 2026-09-15 11:20:00+00 | DEBIT     |  15000.00 | VP1000000001   |
| VP-TXN-20260901-001 | TOPUP    | SUCCESS | 2026-09-01 09:00:00+00 | CREDIT    | 100000.00 | VP1000000001   |
| VP-OB-VP1000000001  | TOPUP    | SUCCESS | 2026-08-25 10:00:00+00 | CREDIT    | 160000.00 | VP1000000001   |
+---------------------+----------+---------+------------------------+-----------+-----------+----------------+
(4 rows)
```

Q2. Daily successful transfer volume

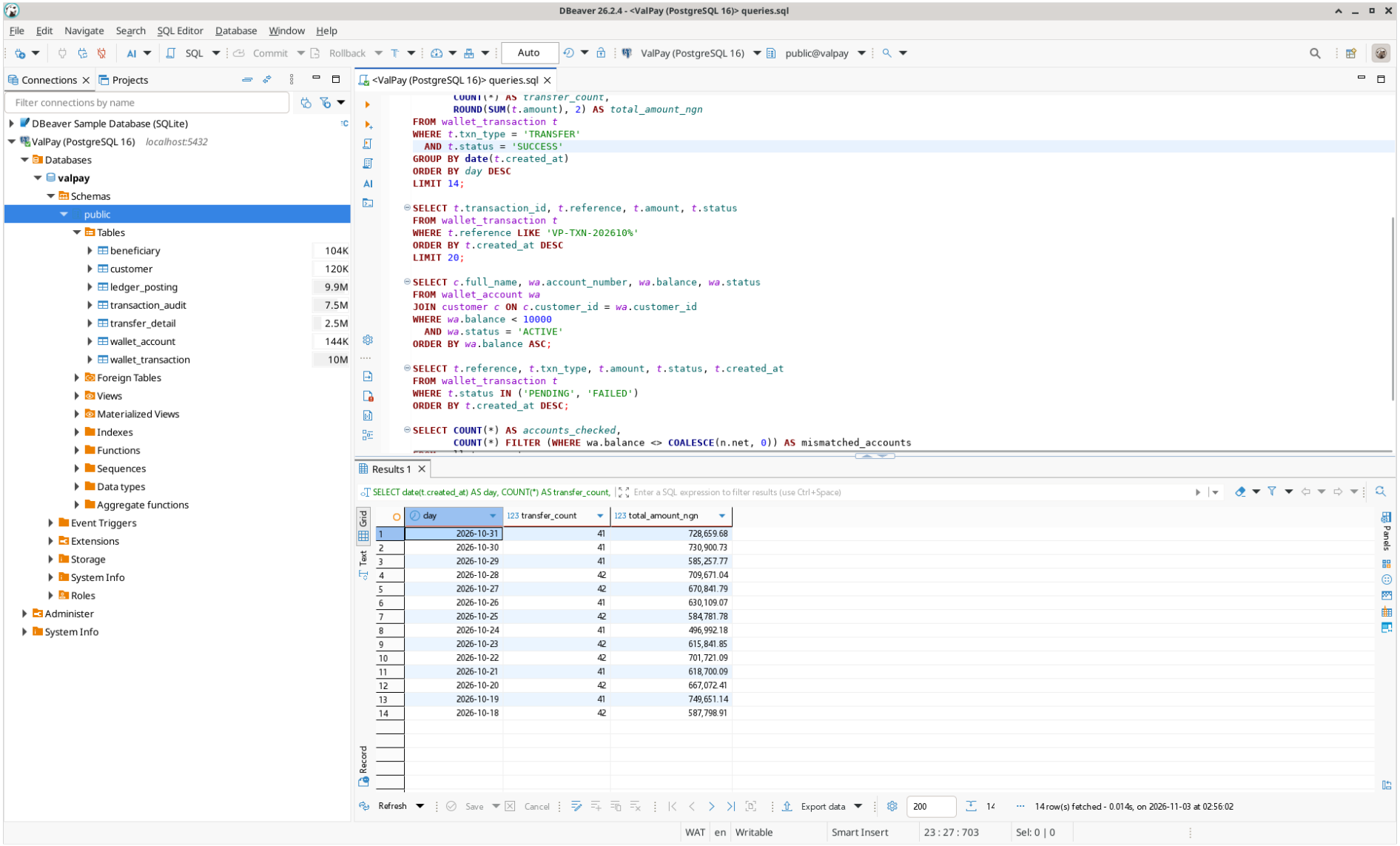


Figure 2. *Q2 in DBeaver: successful transfer count and total amount per day for the latest 14 days.*

Source: evidence/script-output/query_results.txt

```
=== Q2: Daily successful transfer volume (latest 14 days) ===
+------------+----------------+------------------+
|    day     | transfer_count | total_amount_ngn |
+------------+----------------+------------------+
| 2026-10-31 |             41 |        728659.68 |
| 2026-10-30 |             41 |        730900.73 |
| 2026-10-29 |             41 |        585257.77 |
| 2026-10-28 |             42 |        709671.04 |
| 2026-10-27 |             42 |        670841.79 |
| 2026-10-26 |             41 |        630109.07 |
| 2026-10-25 |             42 |        584781.78 |
| 2026-10-24 |             41 |        496992.18 |
| 2026-10-23 |             42 |        615841.85 |
| 2026-10-22 |             42 |        701721.09 |
| 2026-10-21 |             41 |        618700.09 |
| 2026-10-20 |             42 |        667072.41 |
| 2026-10-19 |             41 |        749651.14 |
| 2026-10-18 |             42 |        587798.91 |
+------------+----------------+------------------+
(14 rows)
```

Q3. Lookup by reference prefix

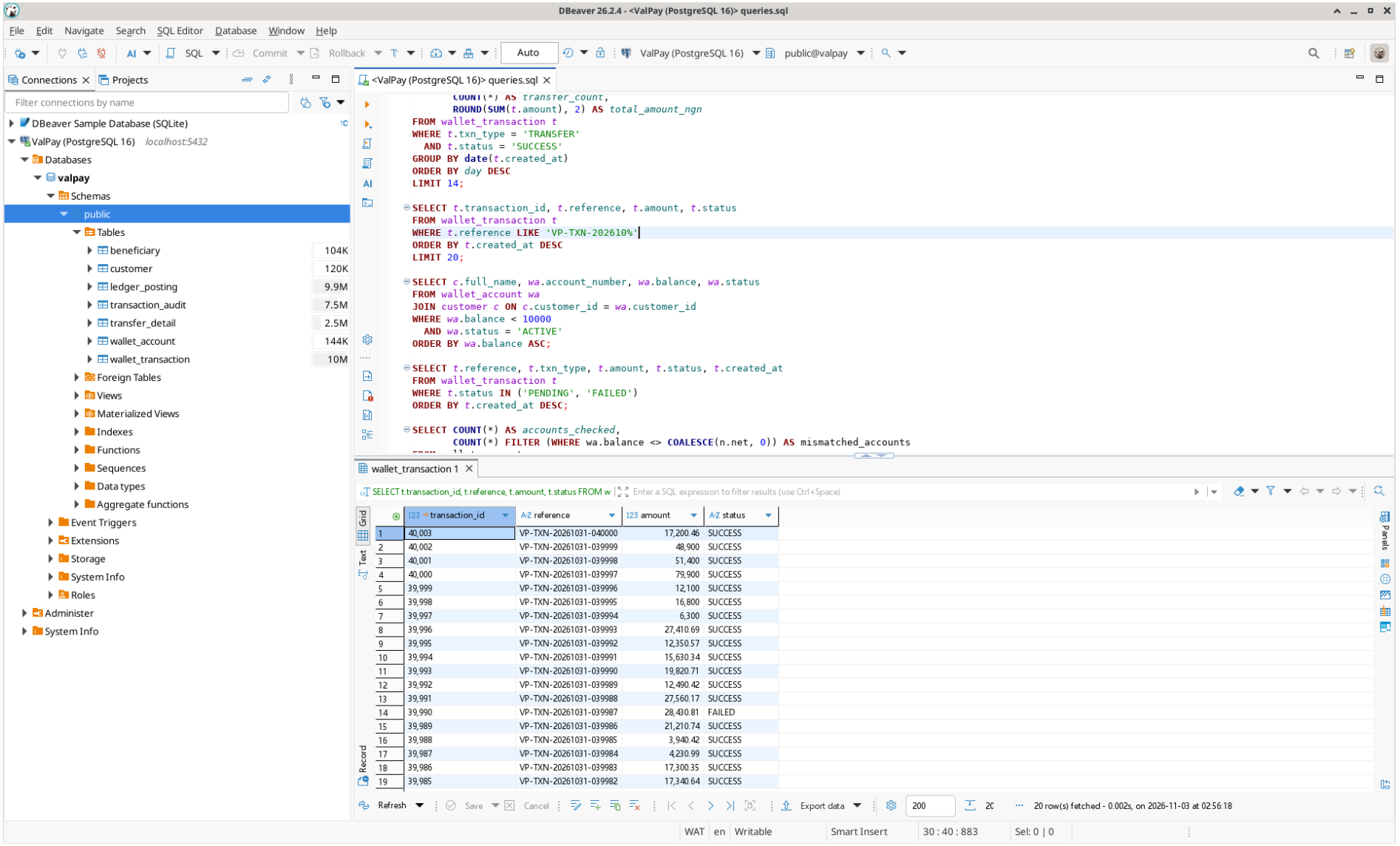


Figure 3. *Q3 in DBeaver: latest 20 transactions whose reference starts with VP-TXN-202610.*

Source: evidence/script-output/query_results.txt

```
=== Q3: Lookup transaction by reference prefix (optimisation candidate, latest 20) ===
+----------------+------------------------+----------+---------+
| transaction_id |       reference        |  amount  | status  |
+----------------+------------------------+----------+---------+
|          40003 | VP-TXN-20261031-040000 | 17200.46 | SUCCESS |
|          40002 | VP-TXN-20261031-039999 | 48900.00 | SUCCESS |
|          40001 | VP-TXN-20261031-039998 | 51400.00 | SUCCESS |
|          40000 | VP-TXN-20261031-039997 | 79900.00 | SUCCESS |
|          39999 | VP-TXN-20261031-039996 | 12100.00 | SUCCESS |
|          39998 | VP-TXN-20261031-039995 | 16800.00 | SUCCESS |
|          39997 | VP-TXN-20261031-039994 |  6300.00 | SUCCESS |
|          39996 | VP-TXN-20261031-039993 | 27410.69 | SUCCESS |
|          39995 | VP-TXN-20261031-039992 | 12350.57 | SUCCESS |
|          39994 | VP-TXN-20261031-039991 | 15630.34 | SUCCESS |
|          39993 | VP-TXN-20261031-039990 | 19820.71 | SUCCESS |
|          39992 | VP-TXN-20261031-039989 | 12490.42 | SUCCESS |
|          39991 | VP-TXN-20261031-039988 | 27560.17 | SUCCESS |
|          39990 | VP-TXN-20261031-039987 | 28430.81 | FAILED  |
|          39989 | VP-TXN-20261031-039986 | 21210.74 | SUCCESS |
|          39988 | VP-TXN-20261031-039985 |  3940.42 | SUCCESS |
|          39987 | VP-TXN-20261031-039984 |  4230.99 | SUCCESS |
|          39986 | VP-TXN-20261031-039983 | 17300.35 | SUCCESS |
|          39985 | VP-TXN-20261031-039982 | 17340.64 | SUCCESS |
|          39984 | VP-TXN-20261031-039981 | 23290.18 | SUCCESS |
+----------------+------------------------+----------+---------+
(20 rows)
```

Q4. Accounts below the monitoring threshold

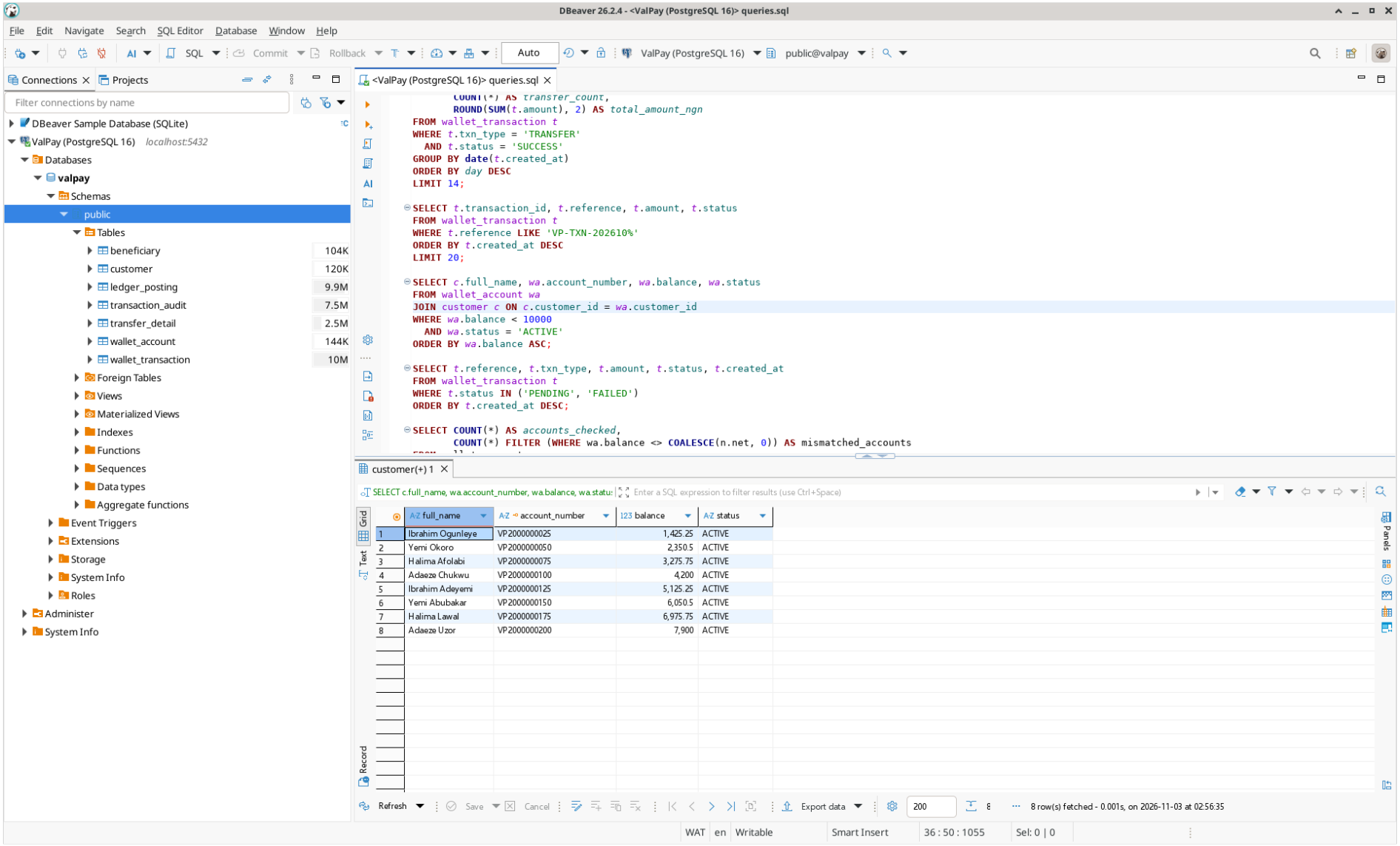


Figure 4. *Q4 in DBeaver: active accounts with a balance below the internal threshold.*

Source: evidence/script-output/query_results.txt

```
=== Q4: Accounts below internal monitoring threshold ===
+------------------+----------------+---------+--------+
|    full_name     | account_number | balance | status |
+------------------+----------------+---------+--------+
| Ibrahim Ogunleye | VP2000000025   | 1425.25 | ACTIVE |
| Yemi Okoro       | VP2000000050   | 2350.50 | ACTIVE |
| Halima Afolabi   | VP2000000075   | 3275.75 | ACTIVE |
| Adaeze Chukwu    | VP2000000100   | 4200.00 | ACTIVE |
| Ibrahim Adeyemi  | VP2000000125   | 5125.25 | ACTIVE |
| Yemi Abubakar    | VP2000000150   | 6050.50 | ACTIVE |
| Halima Lawal     | VP2000000175   | 6975.75 | ACTIVE |
| Adaeze Uzor      | VP2000000200   | 7900.00 | ACTIVE |
+------------------+----------------+---------+--------+
(8 rows)
```

Q5. Pending or failed transactions

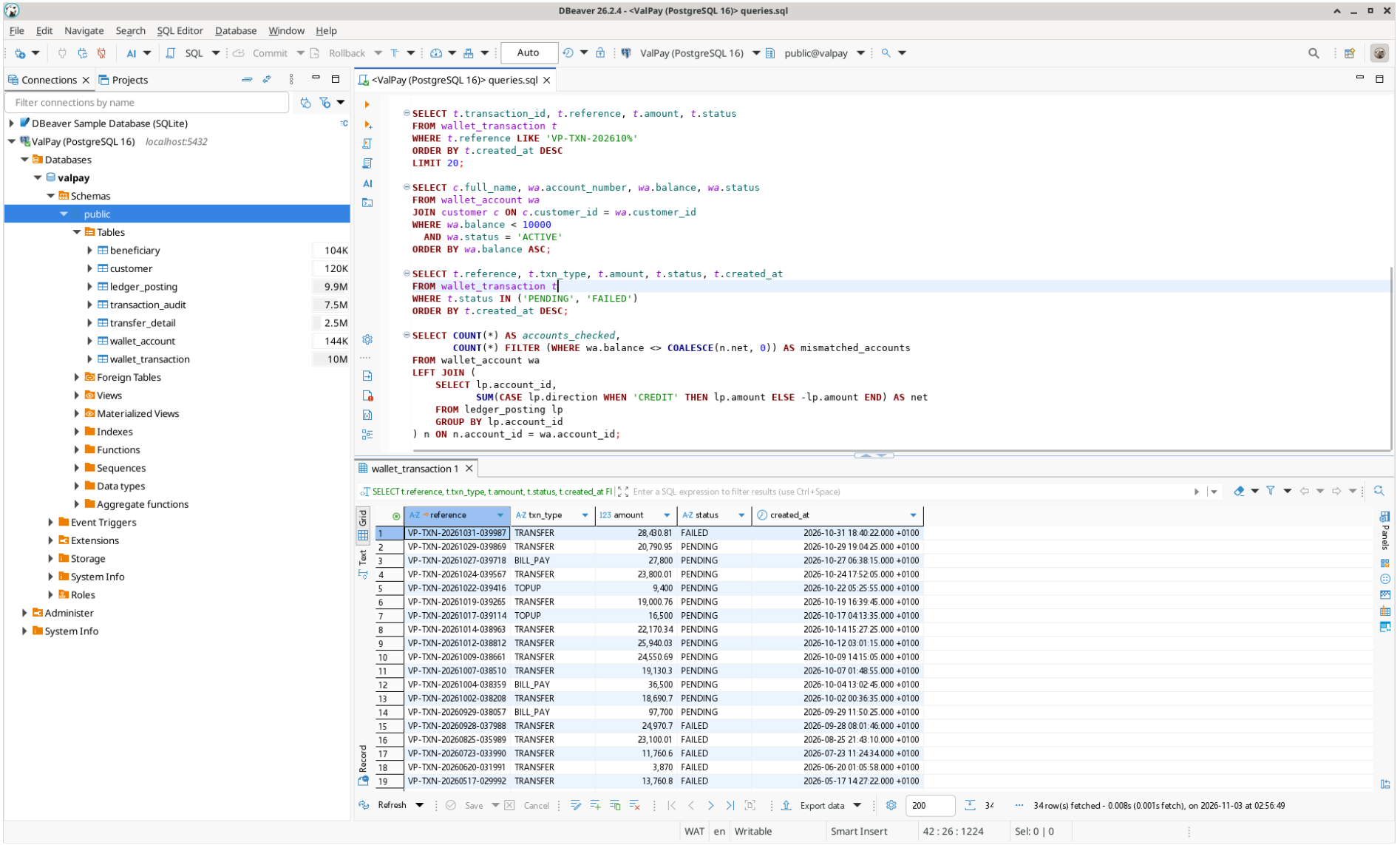


Figure 5. *Q5 in DBeaver: PENDING and FAILED transactions for operations review.*

Source: evidence/script-output/query_results.txt

```
=== Q5: Pending or failed transactions needing ops review ===
+------------------------+----------+----------+---------+------------------------+
|       reference        | txn_type |  amount  | status  |       created_at       |
+------------------------+----------+----------+---------+------------------------+
| VP-TXN-20261031-039987 | TRANSFER | 28430.81 | FAILED  | 2026-10-31 18:40:22+00 |
| VP-TXN-20261029-039869 | TRANSFER | 20790.95 | PENDING | 2026-10-29 19:04:25+00 |
| VP-TXN-20261027-039718 | BILL_PAY | 27800.00 | PENDING | 2026-10-27 06:38:15+00 |
| VP-TXN-20261024-039567 | TRANSFER | 23800.01 | PENDING | 2026-10-24 17:52:05+00 |
| VP-TXN-20261022-039416 | TOPUP    |  9400.00 | PENDING | 2026-10-22 05:25:55+00 |
| VP-TXN-20261019-039265 | TRANSFER | 19000.76 | PENDING | 2026-10-19 16:39:45+00 |
| VP-TXN-20261017-039114 | TOPUP    | 16500.00 | PENDING | 2026-10-17 04:13:35+00 |
| VP-TXN-20261014-038963 | TRANSFER | 22170.34 | PENDING | 2026-10-14 15:27:25+00 |
| VP-TXN-20261012-038812 | TRANSFER | 25940.03 | PENDING | 2026-10-12 03:01:15+00 |
| VP-TXN-20261009-038661 | TRANSFER | 24550.69 | PENDING | 2026-10-09 14:15:05+00 |
| VP-TXN-20261007-038510 | TRANSFER | 19130.30 | PENDING | 2026-10-07 01:48:55+00 |
| VP-TXN-20261004-038359 | BILL_PAY | 36500.00 | PENDING | 2026-10-04 13:02:45+00 |
| VP-TXN-20261002-038208 | TRANSFER | 18690.70 | PENDING | 2026-10-02 00:36:35+00 |
| VP-TXN-20260929-038057 | BILL_PAY | 97700.00 | PENDING | 2026-09-29 11:50:25+00 |
| VP-TXN-20260928-037988 | TRANSFER | 24970.70 | FAILED  | 2026-09-28 08:01:46+00 |
| VP-TXN-20260825-035989 | TRANSFER | 23100.01 | FAILED  | 2026-08-25 21:43:10+00 |
| VP-TXN-20260723-033990 | TRANSFER | 11760.60 | FAILED  | 2026-07-23 11:24:34+00 |
| VP-TXN-20260620-031991 | TRANSFER |  3870.00 | FAILED  | 2026-06-20 01:05:58+00 |
| VP-TXN-20260517-029992 | TRANSFER | 13760.80 | FAILED  | 2026-05-17 14:27:22+00 |
| VP-TXN-20260414-027993 | TRANSFER | 17430.69 | FAILED  | 2026-04-14 04:08:46+00 |
| VP-TXN-20260311-025994 | TOPUP    | 12100.00 | FAILED  | 2026-03-11 17:50:10+00 |
| VP-TXN-20260206-023995 | TOPUP    |  5500.00 | FAILED  | 2026-02-06 07:11:34+00 |
| VP-TXN-20260103-021996 | TOPUP    | 18300.00 | FAILED  | 2026-01-03 20:52:58+00 |
| VP-TXN-20251201-019997 | BILL_PAY | 40000.00 | FAILED  | 2025-12-01 10:34:22+00 |
| VP-TXN-20251029-017998 | BILL_PAY | 47700.00 | FAILED  | 2025-10-29 00:15:46+00 |
| VP-TXN-20250925-015999 | BILL_PAY | 80200.00 | FAILED  | 2025-09-25 13:37:10+00 |
| VP-TXN-20250823-014000 | TRANSFER | 10750.84 | FAILED  | 2025-08-23 03:18:34+00 |
| VP-TXN-20250720-012001 | TRANSFER | 19210.64 | FAILED  | 2025-07-20 16:59:59+00 |
| VP-TXN-20250617-010002 | TRANSFER | 11920.65 | FAILED  | 2025-06-17 06:41:23+00 |
| VP-TXN-20250514-008003 | TRANSFER | 24210.62 | FAILED  | 2025-05-14 20:02:47+00 |
| VP-TXN-20250411-006004 | TRANSFER | 25690.33 | FAILED  | 2025-04-11 09:44:11+00 |
| VP-TXN-20250308-004005 | TRANSFER | 28290.37 | FAILED  | 2025-03-08 23:25:35+00 |
| VP-TXN-20250203-002006 | TRANSFER | 14450.94 | FAILED  | 2025-02-03 13:06:59+00 |
| VP-TXN-20250101-000007 | TRANSFER | 19930.99 | FAILED  | 2025-01-01 02:28:23+00 |
+------------------------+----------+----------+---------+------------------------+
(34 rows)
```

Q6. Ledger reconciliation

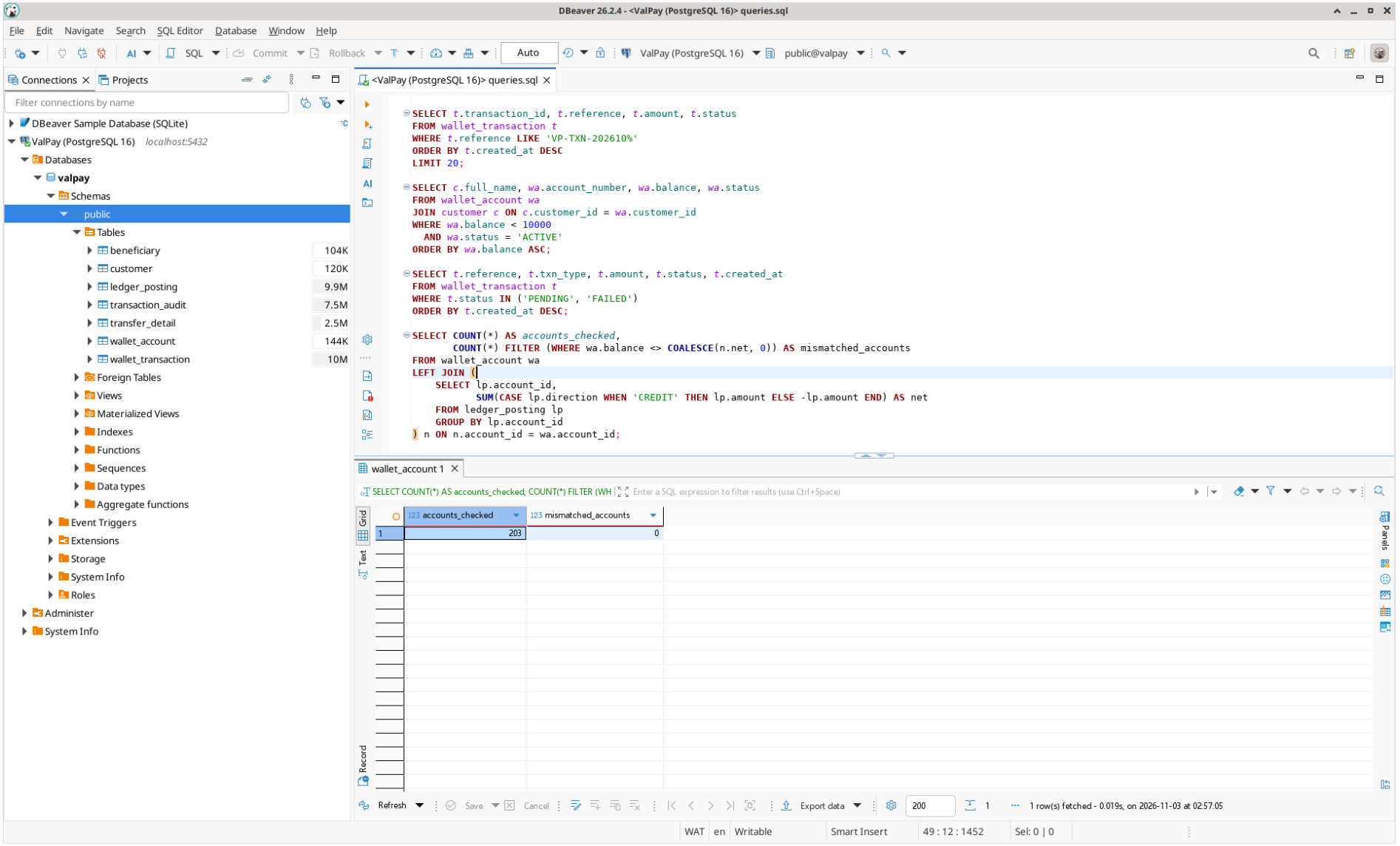


Figure 6. *Q6 in DBeaver: accounts checked against CREDIT minus DEBIT postings, with the mismatch count.*

Source: evidence/script-output/query_results.txt

```
=== Q6: Ledger reconciliation (balance must equal CREDIT - DEBIT postings) ===
+------------------+---------------------+
| accounts_checked | mismatched_accounts |
+------------------+---------------------+
|              203 |                   0 |
+------------------+---------------------+
(1 row)
```

2. EXECUTION PLANS

Q3 before idx_transaction_reference

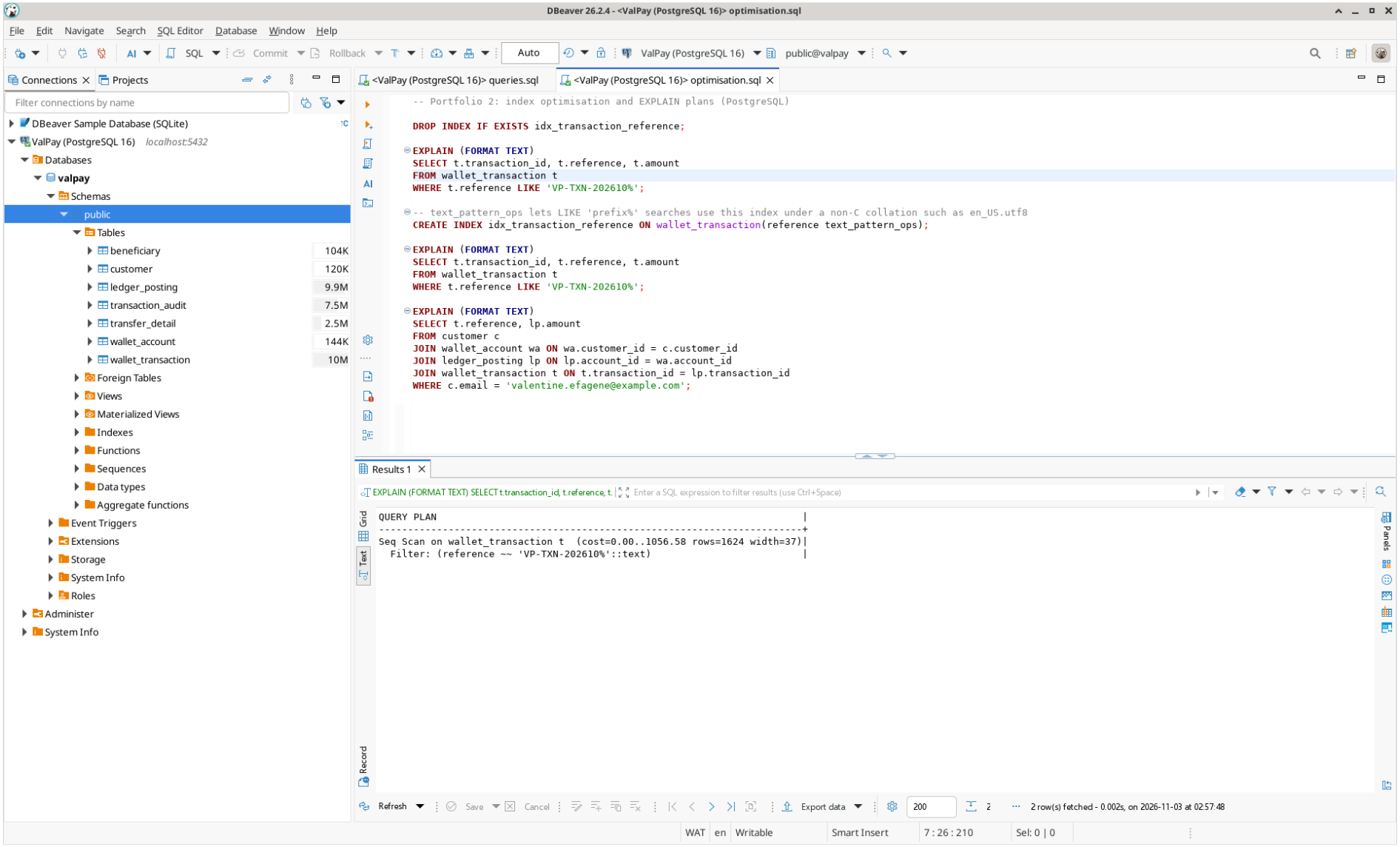


Figure 7. *Q3 before the index: sequential scan on wallet_transaction.*

Source: evidence/script-output/explain_plans.txt

```
DROP INDEX
--- EXPLAIN Q3 BEFORE (no idx_transaction_reference) ---
                                QUERY PLAN
---------------------------------------------------------------------------
 Seq Scan on wallet_transaction t  (cost=0.00..1056.58 rows=1624 width=37)
   Filter: (reference ~~ 'VP-TXN-202610%'::text)
(2 rows)

CREATE INDEX
```

Q3 after idx_transaction_reference

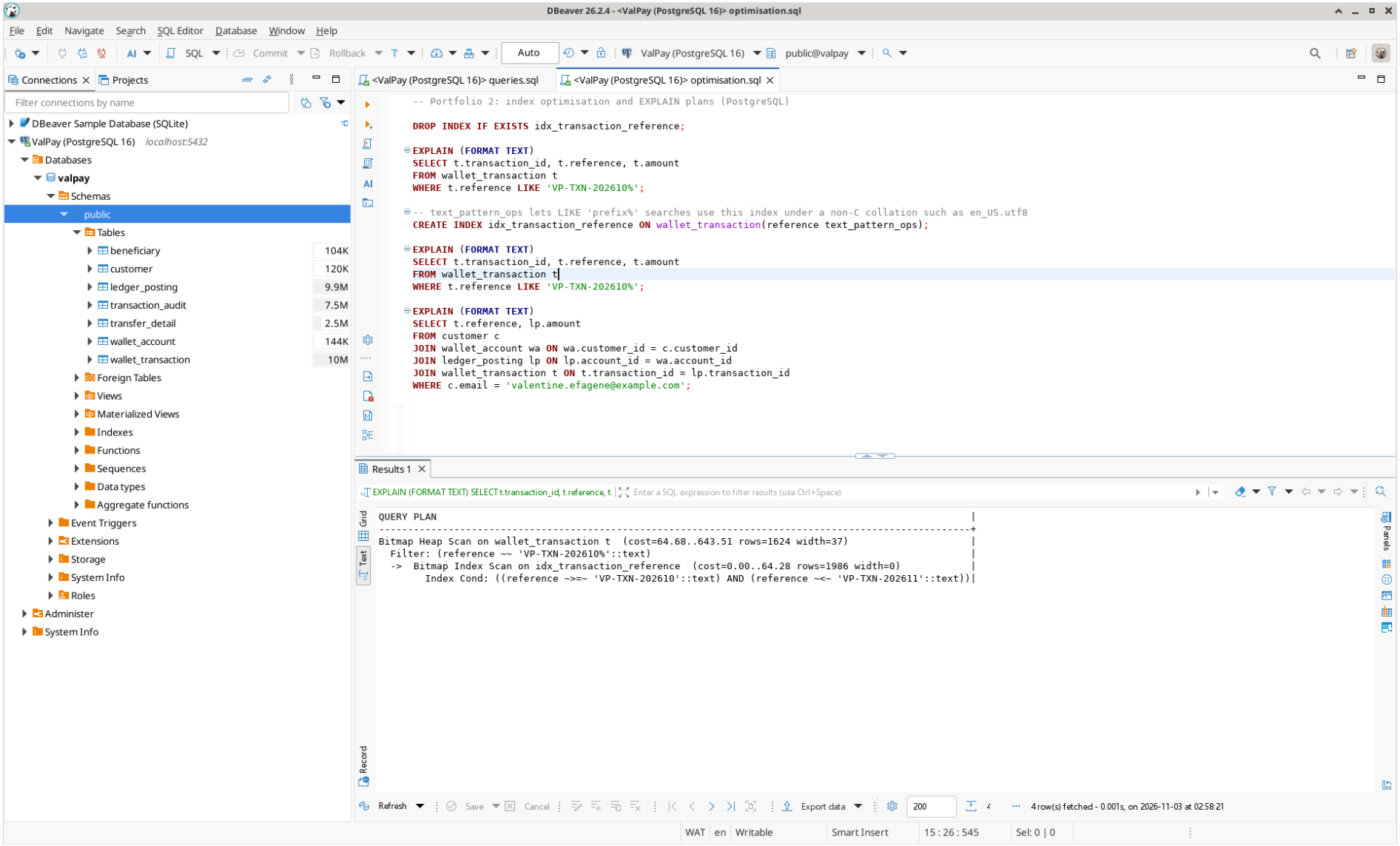


Figure 8. *Q3 after the index: bitmap index scan on idx_transaction_reference.*

Source: evidence/script-output/explain_plans.txt

```
--- EXPLAIN Q3 AFTER (idx_transaction_reference) ---
                                                 QUERY PLAN
-------------------------------------------------------------------------------------------------------------
 Bitmap Heap Scan on wallet_transaction t  (cost=64.68..643.51 rows=1624 width=37)
   Filter: (reference ~~ 'VP-TXN-202610%'::text)
   ->  Bitmap Index Scan on idx_transaction_reference  (cost=0.00..64.28 rows=1986 width=0)
         Index Cond: ((reference ~>=~ 'VP-TXN-202610'::text) AND (reference ~<~ 'VP-TXN-202611'::text))
(4 rows)
```

Q1 join plan

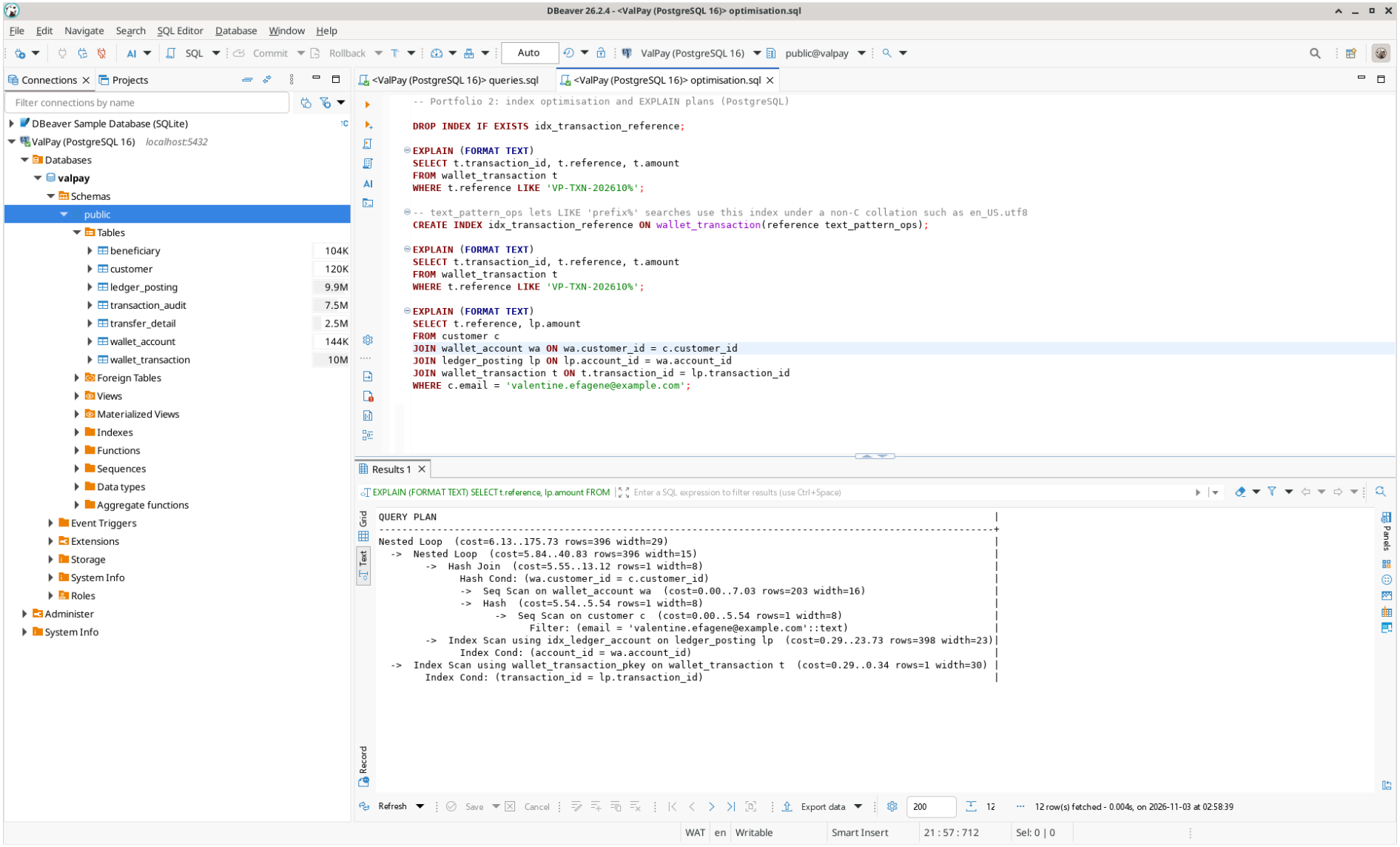


Figure 9. *Q1 plan: nested loops using idx_ledger_account and wallet_transaction_pkey.*

Source: evidence/script-output/explain_plans.txt

```
--- EXPLAIN Q1 (ledger by account via join) ---
                                                 QUERY PLAN
-------------------------------------------------------------------------------------------------------------
 Nested Loop  (cost=6.13..175.68 rows=396 width=29)
   ->  Nested Loop  (cost=5.84..40.78 rows=396 width=15)
         ->  Hash Join  (cost=5.55..13.12 rows=1 width=8)
               Hash Cond: (wa.customer_id = c.customer_id)
               ->  Seq Scan on wallet_account wa  (cost=0.00..7.03 rows=203 width=16)
               ->  Hash  (cost=5.54..5.54 rows=1 width=8)
                     ->  Seq Scan on customer c  (cost=0.00..5.54 rows=1 width=8)
                           Filter: (email = 'valentine.efagene@example.com'::text)
         ->  Index Scan using idx_ledger_account on ledger_posting lp  (cost=0.29..23.69 rows=396 width=23)
               Index Cond: (account_id = wa.account_id)
   ->  Index Scan using wallet_transaction_pkey on wallet_transaction t  (cost=0.29..0.34 rows=1 width=30)
         Index Cond: (transaction_id = lp.transaction_id)
(12 rows)
```